1 **(a)** B1 for a smooth U-shaped parabola passing through $(0, 1)$, $(2, -3)$, and $(4, 1)$

B1 for correct endpoints plotted or stopped cleanly at $(-1, 6)$ and $(5, 6)$

(b) $(2, -3)$ **[B1]**

(c) 0.27 **[B1]**

3.73 **[B1]**

[Total Marks: 5]

2 **(a)** Correct inverted U-shape parabola **[B1]**

Maximum approximately at $(3, 4)$ and crosses the x-axis at 1 and 5 **[B1]**

(b) $x = 1$ **[B1]**

$x = 5$ **[B1]**

(c) $(3, 4)$ **[B2]**

If B0, M1 for $x = 3$ or $y = 4$ identified

[Total Marks: 6]

3 **(a)** Correct shape (inverted parabola) **[B1]**

Graph passes approximately through $(-0.5, 0)$, $(3, 0)$, $(0, 3)$ and vertex roughly at $(1.25, 6.1)$ **[B1]**

(b) $(1.25, 6.125)$ (Accept 6.13 or 6.1) **[B2]**

If B0, M1 for identifying correct x-coordinate 1.25 or correct derivative $-4x + 5 = 0$

(c) $x = -0.5$ **[B1]**

$x = 3$ **[B1]**

[Total Marks: 6]

4 **(a)** Correct basic cubic shape with a local maximum and local minimum **[B1]**

Graph crosses y-axis at 2, has a maximum at $(0, 2)$ and a minimum at $(2, -2)$ **[B1]**

(b) Uses GDC to find roots **[M1]**

$-0.732, 1, 2.73$ (accept to 3 s.f.) **[A1]**

(c) Identifies correct turning point from GDC **[M1]**

$(2, -2)$ **[A1]**

[Total Marks: 6]

5 **(a)** Correct U-shaped parabola passing through the y-axis at -3 **[B1]**

Vertices and endpoints approximately in the correct positions **[B1]**

(b) -0.5 **[B1]**

3 **[B1]**

(c) $x = 1.25$ **[B1]**

$y = -6.125$ **[B1]**

[Total Marks: 6]

6 (a) B1 for a smooth cubic curve with y-intercept at $(0, 1)$ and correct general shape

B1 for correct turning points in the second and fourth quadrants and correct x-intercepts near -3.05, 0.52, and 2.53

(b) Local maximum $= (-1.63, 3.18)$ **[B1]**

Local minimum $= (1.63, -1.18)$ **[B1]**

(c) $-1.18 < k < 3.18$ **[B2]**

If 0 scored, allow B1 for identifying the boundary values -1.18 and 3.18

[Total Marks: 6]

7 (a) Correct cubic shape with a local maximum and a local minimum **[B1]**

Local maximum approximately at $(-1, 3)$ and local minimum approximately at $(1, -1)$ **[B1]**

(b) $(-1, 3)$ **[B2]**

If B0, M1 for $x = -1$ or $y = 3$

(c) Correct straight line passing through $(0, -1)$ and $(2, 3)$ **[B1]**

(d) $x = 2$ **[B1]**

$x = 0.414$ and $x = -2.41$ **[B1]**

[Total Marks: 7]

8 (a) Straight line for $g(x)$ passing through $(0, -2)$ and $(2, 0)$ **[B1]**

Two correct hyperbolic branches for $f(x)$ **[B1]**

Correct asymptotic behaviour near $x = -1$ **[B1]**

(b) $x = -1$ **[B1]**

(c) Sets up $\dfrac{2}{x+1} = x - 2$ or uses intersection feature on GDC **[M1]**

One correct coordinate pair: $(-1.56, -3.56)$ or $(2.56, 0.562)$ (to 3 s.f.) **[A1]**

Second correct coordinate pair **[A1]**

[Total Marks: 7]

9 (a) Correct basic cubic shape **[B1]**

Local maximum near $x = 0$, local minimum near $x = 2.5$, and passing through $(0, 2)$ **[B1]**

(b) 1 **[B1]**

-0.562 (or $-0.561...$) **[B1]**

3.56 (or 3.561...) **[B1]**

(c) $x = 2.54$ (or 2.535...) **[B1]**

$y = -4.88$ (or $-4.879...$) **[B1]**

(d) Straight line passing through $(0, 1)$ with a negative gradient **[B1]**

(e) -0.468 (allow -0.47) **[B1]**

0.743 (allow 0.74) **[B1]**

3.73 **[B1]**

[Total Marks: 11]

10 (a) Correct general cubic shape starting bottom left and ending top right **[B1]**

Local maximum correctly positioned at approximately $(-1, 4)$ **[B1]**

Local minimum correctly positioned at $(1, 0)$ **[B1]**

(b) $(-1, 4)$ **[B2]**

If B0, M1 for attempting to find stationary points e.g. evaluating max on calculator or $3x^2 - 3 = 0$

(c) $x = -2$ **[B1]**

$x = 1$ **[B1]**

(d) Straight line with positive gradient **[B1]**

Line passing through $(0, 1)$ and $(-0.5, 0)$ **[B1]**

(e) Uses calculator to find intersections or equates $x^3 - 3x + 2 = 2x + 1$ **[M1]**

-2.33 (or $-2.330\ldots$) **[B1]**

0.202 (or $0.2016\ldots$) and 2.13 (or $2.128\ldots$) **[B2, or B1 for 1 correct]**

[Total Marks: 12]

11 (a) B1 for $y = f(x)$ starting at $(0, 0)$, rising to a single peak just above 1.5, and decaying towards the x-axis

B1 for $y = g(x)$ drawn as a straight line from $(0, 0.2)$ to $(5, 2.7)$

B1 for clearly showing two points of intersection in the first quadrant

(b) $(1.44, 1.59)$ **[B2]**

If 0 scored, allow B1 for $x = 1.44$ or $y = 1.59$

(c) M1 for finding the x-coordinates of the points of intersection: 0.08 (or $0.082\ldots$) and 2.46 (or $2.456\ldots$)

$0.08 < x < 2.46$ **[A1]**

[Total Marks: 7]

12 **(a)** Graph touches the origin $(0, 0)$ **[B1]**

Graph reaches a maximum in the first quadrant and then approaches the x-axis asymptotically **[B1]**

Curve sketched with reasonable accuracy from $x = -1$ to $x = 5$, with $f(-1) = 2$ **[B1]**

(b) Uses maximum feature on GDC **[M1]**

$(2.89, 1.13)$ (accept $x = 2.88$ to 2.89 and $y = 1.12$ to 1.13) **[A1]**

(c) Draws line $y = 0.5$ on GDC or sets up equation **[M1]**

$-0.589, 0.835, 5.38$ (all three required, accept to 3 s.f.) **[A1]**

(d) Recognises the horizontal line $y = k$ must lie between the local minimum (0) and local maximum (1.13) **[M1]**

$0 < k < 1.13$ **[A1]**

[Total Marks: 9]

13 **(a)** Correct W-shaped curve passing through the origin **[B1]**

Two minima (one in the third quadrant, one in the first quadrant) and one maximum roughly in correct places **[B1]**

Graph drawn between $x = -2$ and $x = 3$ with correct relative heights at the endpoints **[B1]**

(b) $(-1, -19)$ **[B1]**

$(2, 8)$ **[B1]**

(c) $(1, 13)$ **[B1]**

(d) Uses the y-values of the local extremum in the first quadrant to set bounds **[M1]**

$8 < k < 13$ **[A1]**

(e) $(1, 18)$ **[B1]**

[Total Marks: 9]

14 **(a)** Correct shape with two distinct branches **[B1]**

Left branch passing through $(0, -2)$ and tending downwards as x

2^- [B1]

Right branch passing through $(4, 6)$ and tending upwards as x

2^+ [B1]

(b) $x = 2$ **[B1]**

(c) $(0, -2)$ **[B2]**

If B0, M1 for either coordinate correct

(d) $(4, 6)$ **[B2]**

If B0, M1 for either coordinate correct

(e) Identifies the critical values -2 and 6 from the local extrema **[M1]**

$-2 < k < 6$ **[A1]**

[Total Marks: 10]

15 (a) Left branch in the 2nd and 3rd quadrants, passing through x-axis and asymptotic to y-axis downwards **[B1]**

Right branch in 1st quadrant with a clear minimum, asymptotic to y-axis upwards **[B1]**

Both branches curving upwards as x

$\pm\infty$ [B1]

(b) $(1, 3)$ **[B2]**

If B0, M1 for evaluating turning point on calculator or setting derivative to 0

(c) -1.26 (or $-1.259\ldots$) **[B1]**

(d) $x = 0$ **[B1]**

(e) Recognises need to find intersections of graph with $y = 4$ or rearranging to $x^3 - 4x + 2 = 0$ **[M1]**

-2.21 (or $-2.214\ldots$) **[B1]**

0.539 (or $0.5391\ldots$) and 1.68 (or $1.675\ldots$) **[B1]**

[Total Marks: 10]